Skills Summary

Adding and Subtracting with Like Denominators

Use this reference while completing your worksheet or when studying.

Skill 1 — Adding with like denominators

Rule: when the denominators match, the pieces are the same size — add the numerators (the counts of pieces) and *keep the denominator*. Simplify the answer if you can.

Example: Add $\frac{2}{5} + \frac{1}{5}$.
Step 1. Both fractions count fifths — same-size pieces — so I can add the counts and keep the denominator.
Step 2. $2 + 1 = 3$ pieces, each one fifth: $\frac{2}{5} + \frac{1}{5} = \mathbf{\frac{3}{5}}$

Try it: Add $\frac{3}{7} + \frac{2}{7}$.

check: $\mathbf{\frac{5}{7}}$

Skill 2 — Subtracting with like denominators

Rule: subtract the numerators and keep the denominator. Taking pieces away never changes the size of the pieces.

Example: Subtract $\frac{7}{10} - \frac{3}{10}$.
Step 1. Both fractions count tenths, so subtract the counts and keep the denominator.
Step 2. $7 - 3 = 4$ tenths, and $\frac{4}{10}$ simplifies (divide top and bottom by 2) to $\mathbf{\frac{2}{5}}$

Try it: Subtract $\frac{5}{9} - \frac{2}{9}$. Simplify your answer.

check: $\mathbf{\frac{1}{3}}$

Skill 3 — Breaking a fraction into parts (decomposing)

Idea: every fraction is a sum of smaller fractions with the same denominator. You can always break it into unit fractions: $\frac{3}{4} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$.

Example: Fill in: $\frac{4}{6} = \frac{3}{6} + \frac{\square}{6}$
Step 1. The two parts must add back to 4 sixths, so ask: 3 plus what makes 4?
Step 2. $4 - 3 = 1$, so the missing numerator is **1**

Try it: Fill in: $\frac{7}{8} = \frac{5}{8} + \frac{\square}{8}$

check: **2**

(!) Watch out: never add the denominators. $\frac{2}{5} + \frac{1}{5}$ is *not* $\frac{3}{10}$ — the denominator names the size of the pieces, and the pieces stay fifths.